STAT 516 MIDTERM
40 pt total

1. (10pt) A die is rolled twice. Let us assume that all the elementary events in the sample space $\mathcal{S} = \{(i, j) : i, j = 1, 2, \ldots, 6\}$ are equally likely. Define A as the event that the first throw shows a number on the top face ≤ 2, and B, the event that the second throw shows at least 5.

   (a) (5pt) What is the probability $P(A \cap B)$?

   (b) (5pt) What is the probability $P(A \cup B)$?

   Solution: Clearly, $A = \{(i, j) : 1 \leq i \leq 2, j = 1, 2, \ldots, 6\}$ and $B = \{(i, j) : 5 \leq j \leq 6, i = 1, 2, \ldots, 6\}$. Then, $P(A \cap B) = P(\{(1, 5), (1, 6), (2, 5), (2, 6)\}) = \frac{4}{36} = \frac{1}{9}$. Then, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{3} + \frac{1}{3} - \frac{1}{9} = \frac{5}{9}$

2. (10pt) One of the two urns has a red and b black balls, and the other has c red and d black balls. One ball is chosen at random from each urn, and then one of these two balls is chosen at random. What is the probability that this ball is red?

   Solution: if each ball selected from the two urns is red, then the final ball is definitely red. If one of these two balls is red, then the final balls is red with probability $\frac{1}{2}$. If none of those two balls is read, then the final ball cannot be red.

   Thus, $P(the final ball is red) = \frac{a}{a+b} \times \frac{c}{c+d} + \frac{1}{2} \times \left(\frac{a}{a+b} \times \frac{d}{c+d} + \frac{b}{a+b} \times \frac{c}{c+d} \right) = \frac{2ac+ad+bc}{2(a+b)(c+d)}$

3. (10pt) Define the pdf of the standard Laplace (double exponential distribution) as $f(x) = \frac{1}{2}e^{-|x|}$, $-\infty < x < \infty$

   (a) (5pt) Show that the function $f(x)$ is, indeed, a pdf

   (b) (5pt) Find its cdf

   Solution: directly by integration, the cdf is $F(x) = e^{x}/2$ for $x \le 0$, and $F(x) = 1 - e^{-x}/2$ if $x > 0$.

4. (10pt) Let X be the sum of the two rolls when a fair die is rolled twice.

   (a) (5pt) Write down a pmf of X. You can do it in the form of a table, if you like.

   (b) (5pt) Find $\mathbb{E}\,X$

   Solution: the pmf of X is $p(2) = p(12) = 1/36$; $p(3) = p(11) = 2/36 = 1/18$; $p(4) = p(10) = 3/36 = 1/12$; $p(5) = p(9) = 4/36 = 1/9$;$p(6) = p(8) = 5/36$;$p(7) = 6/36 = 1/6$. Thus, $\mathbb{E}\,X = 2 \times 1/36 + 3 \times 2/36 + 4 \times 3/36 + \cdots + 12 \times 1/36 = 7$. Alternatively, if X_1=the face obtained on the first roll, X_2=the face obtained on the second roll, $\mathbb{E}\,X = \mathbb{E}(X_1 + X_2) = 2 \times 3.5 = 7$